

FRAME: Feature Rectification for Class Imbalance Learning

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Abstract—Class imbalance learning is a challenging task in machine learning applications. To balance training data, traditional class imbalance learning approaches, such as class resampling or reweighting, are commonly applied in the literature. However, these methods can have significant limitations, particularly in the presence of noisy data, missing values, or when applied to advanced learning paradigms like semi-supervised or federated learning. To address these limitations, this paper proposes a novel and theoretically-ensured latent Feature Rectification method for class imbalance learning (FRAME). The proposed FRAME can automatically learn multiple centroids for each class in the latent space and then perform class balancing. Unlike data-level methods, FRAME balances feature in the latent space rather than the original space. Compared to algorithm-level methods, FRAME can distinguish different classes based on distance without the need to adjust the learning algorithms. Through latent feature rectification, FRAME can effectively mitigate contaminated noises/missing values without worrying about structural variations in the data. In order to accommodate a wider range of applications, this paper extends FRAME to the following three main learning paradigms: fully-supervised learning, semi-supervised learning, and federated learning. Extensive experiments on 10 binary-class datasets demonstrate that our FRAME can achieve competitive performance than the state-of-the-art methods and its robustness to noises/missing values.

Index Terms—Class imbalance learning, feature rectification, latent space, learning paradigm.

I. INTRODUCTION

CLASS imbalance is a popular subject in machine learning since the potential assumption of machine learning is that the model should be trained on nearly balanced samples between classes [1]. However, this assumption is not always

true in the real world, as the imbalance between classes can be widely observed [2]. For example, in building a machine learning model for cancer detection, the number of samples for healthy people is much higher than that of “ill” people. Faced with the class imbalance problem, conventional machine learning models usually form the so-called “majority bias”. ‘Majority bias’ refers to the tendency of machine learning models to prioritize or favor the majority class over the minority class in imbalanced datasets. This occurs because the model learns to minimize overall error, which can be achieved more easily by correctly predicting the majority class, often at the expense of the minority class. This result is sub-optimal because the information from a few majority classes is always concerned.

To mitigate the issue of class imbalance, a variety of traditional methods have been developed, classified into three main groups: data-level, algorithm-level, and ensemble methods [3]. Data-level techniques address the imbalance by either increasing the instances of the minority class or decreasing those of the majority class [4]. Algorithm-level techniques adjust the learning process to more effectively recognize features of the minority class [5]. Ensemble methods combine the power of ensemble learning algorithms to integrate the benefits of the earlier approaches into a coherent classifier [6].

Data-level strategies seek to equalize the distribution of training samples and encompass methods such as over-sampling, under-sampling, and hybrid sampling techniques [4]. While they have proven to be effective, these techniques are computationally demanding for large-scale datasets and inadequate for datasets with missing values because they depend on distance metrics. Algorithm-level approaches address ‘majority bias’ by modifying existing learning algorithms to impose higher penalties on the minority class [6]. Even though these methods are effective, they necessitate the creation of task-specific cost matrices and are impractical for batch-trained neural networks [7]. Ensemble approaches, like those detailed by Yu et al. [8] and Zhang et al. [9], show significant benefits in managing noisy data by using an array of models to improve robustness. However, they also bring certain challenges; ensembles can be computationally demanding in environments with limited resources and may show decreased sensitivity to noise in cases of high-dimensional data or highly intricate noise distributions [10].

Recently, there has been growing interest in combining deep representation learning with class imbalance techniques. In such methods, balancing operations are performed within the latent space [11], which represents a lower-dimensional

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representation where a neural network captures the most relevant and informative features of the original data. The core belief is that latent features provide better compactness and separability than raw data [12]. Nevertheless, these methods based on deep latent features face challenges. First, the latent space may be affected by data noise, outliers, or missing values, leading to a dispersion of intra-class samples. Second, current research ignores the different levels of importance among samples within the same class, failing to recognize that samples nearer to the latent space center hold more significance.

To address the aforementioned challenges, this paper introduces a novel latent **Feature Rectification** method for **clAss iMbalance lEarning** (FRAME). Initially, raw class-imbalance data is projected into a latent feature space. Subsequently, FRAME autonomously learns multiple centroids for each class within this partitioned latent space. The key contributions of this work are:

- FRAME offers a theoretically-grounded latent space feature rectification module for class imbalance learning. Unlike data-level methods that balance features in the original data space, FRAME operates in the latent space. In contrast to algorithm-level methods, FRAME classifies based on distance metrics, eliminating the need for algorithmic adjustments. This latent space rectification allows FRAME to effectively neutralize noise and missing values in the raw data without altering the data structure.
- FRAME is highly adaptable, integrating smoothly into various class imbalance learning paradigms such as fully-supervised, semi-supervised, and federated learning. We conduct comprehensive experiments on 20 public binary-class imbalance datasets across diverse domains, with imbalance ratios ranging from 2.46 to 578.88, as well as two large-scale multi-class datasets. Experimental results demonstrate that FRAME consistently outperforms or matches state-of-the-art algorithms in fully-supervised, semi-supervised, and federated learning paradigms.

The remainder of this paper is structured as follows. Section II gives an overview of class imbalance learning, and surveys the methods for class imbalance learning; The proposed method is described in Section III; Section IV discusses the results; Section V concludes the paper, and presents the future research directions.

II. RELATED WORKS

This section provides an overview of the existing methods for class imbalance learning, which can be broadly classified into three categories: conventional class imbalance learning methods and latent feature balancing based methods. We highlight the pros and cons of these methods and compare them to our proposed FRAME module.

A. Class Imbalance Learning Methods

In the realm of machine learning, addressing class imbalance encompasses a variety of approaches, which can be categorized into data-level, algorithmic, and ensemble methods. Data-level

strategies involve both oversampling and undersampling to balance class distributions. Techniques such as SMOTE [13] and its variants [14], along with noise-resilient [15], weakly supervised [16], and clustering-based oversampling [17], are used to generate higher-quality synthetic samples and mitigate overfitting issues. In contrast, undersampling strategies aim to retain the crucial information from the majority class while decreasing its size, employing methods like metaheuristic techniques [18] and neighborhood analysis [19].

Algorithmic strategies tackle class imbalance by altering the learning process. Cost-sensitive learning assigns varying costs to classification mistakes, with a specific focus on the minority class to ensure the model's balanced attention [4], or it can function at the data level by allocating costs to each instance. Active learning, another crucial technique, identifies and selects the most informative and relevant examples from a set of unlabeled data, thereby optimizing model performance and efficiency. Both methods aim to improve learning efficacy by concentrating on the most significant aspects of the data [20].

Ensemble techniques combine multiple models to improve prediction accuracy and handle class imbalances. These range from conventional methods like bagging and boosting to advanced techniques that incorporate GANs and cost-sensitive elements [21], [22]. The FRAME module, introduced in this research, innovatively combines data-level and algorithmic approaches. It seeks to overcome the limitations of individual strategies by offering a comprehensive solution that enhances machine learning models' performance in handling class imbalances, marking a significant advancement in the field.

B. Latent Feature Balancing Based Methods

Recent research has shifted focus towards balancing feature distributions in the latent space, rather than the data space. This approach, termed feature rectification in the latent space, is utilized for objectives like domain adaptation [23] and class imbalance learning. For instance, deep oversampling techniques generate new samples in the latent space using CNNs [11], while other studies propose methods like batch-size incremental hard sample mining [24] and class rectification loss specifically for minority classes. Approaches such as the calculation of mean feature representation for each class [25], integrating deep learning with k-means clustering [12], and hybrid loss functions incorporating classification [26] and clustering have also been explored [27]. However, these methods face limitations such as susceptibility to data noise, requirement of large datasets, and dependence on hyperparameters.

Building on these foundations, our research introduces the FRAME model for class imbalance learning. This model distinguishes itself by fully leveraging the latent space structure, emphasizing the importance of each sample, and employing attention and metric embedding techniques to learn multiple centroids for each class. Unlike previous methods, FRAME is designed to be generic and theoretically robust, offering a comprehensive solution to the challenges of class imbalance in both supervised and other learning paradigms.

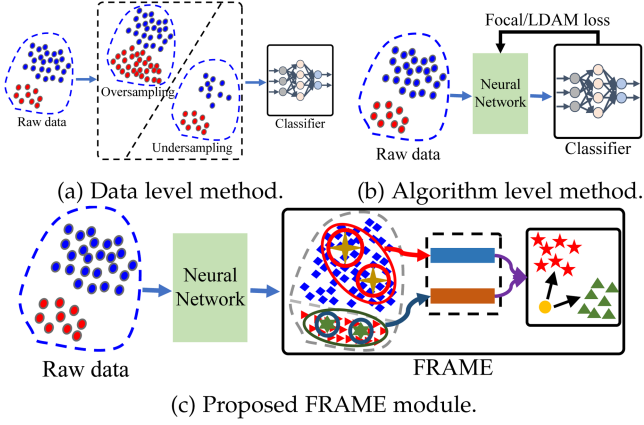


Fig. 1. Comparison between the existing methods and the proposed FRAME module. Unlike the data-level method and algorithm-level method that directly operate raw data or control model training, our proposed FRAME module modulates the imbalanced features in the latent space, and then a distance-based classifier is built for balanced learning.

III. FRAME

In this section, we begin by defining the problem of class-imbalanced classification. Next, we introduce our FRAME module, including its key components and theoretical proof for its effectiveness in addressing class-imbalanced learning. Finally, we demonstrate how the FRAME module can be adapted to various learning paradigms for class imbalance problems, including fully-supervised learning, semi-supervised learning, and federated learning.

A. Problem Definition

Without losing generality, we define imbalanced binary classification as our research problem, and the proposed method can be extended to multi-class settings. In the training samples, there are two classes: the majority class (denoted by $\mathcal{N} = \{(x^N, y^N) \mid y = 0\}$) and the minority class (denoted by $\mathcal{P} = \{(x^P, y^P) \mid y = 1\}$). For an imbalance classification problem, the imbalance ratio (IR) is used to define the uniform level between the two classes: $IR = \frac{|\mathcal{N}|}{|\mathcal{P}|}$, where $|\cdot|$ represents the size of the sample set, $|\mathcal{N}| \gg |\mathcal{P}|$.

The goal of class imbalance learning is to address the issue of imbalanced datasets in machine learning, where the distribution of classes in the dataset is imbalanced. This can be a problem because machine learning algorithms are typically designed to optimize for accuracy, and when there is a large imbalance in the number of examples in each class, the algorithm may focus disproportionately on the majority class at the expense of the minority class. As a result, the minority class may be poorly represented or classified, leading to poor performance overall. In this work, we try to improve the learning performance of both classes by using feature rectification in the latent space.

B. FRAME Module

1) *Overview*: As depicted in Fig. 1(a), the data-level method operates directly on raw data within the original data space. However, this approach necessitates explicit label information,

rendering it unsuitable for learning paradigms that lack labeled samples, such as semi-supervised and weakly supervised learning. In contrast, the algorithm-level method (Fig. 1(b)) governs the model training process and can achieve competitive results by employing specially designed loss functions. However, this method often requires extensive hyperparameter tuning. Our proposed method aims to address class imbalance within the latent space. Fig. 1(c) provides an overview of the FRAME module, which comprises two sequential components: centroid learning to balance imbalanced latent features and a distance-based classifier that calculates the probability of a sample belonging to a particular class based on its distance from the learned centroids.

Data collected from real-world settings often exhibit missing values and noise due to sensor failures or communication issues. Directly oversampling or downsampling this unprocessed data can introduce further noise into subsequent modeling stages, resulting in decreased model performance. Conversely, the proposed FRAME method initially converts the incomplete and noisy raw data into latent space via a parametric mapping, referred to as $\Phi: \mathcal{X} \rightarrow \mathcal{F}$. Following this, multiple centroids for each class are calculated to ensure feature balance. These centroids act as the weighted average of the derived features, reducing feature noise within the latent space.

Various advantages stem from employing centroids for class representation: first, they provide a computational benefit due to the low complexity of centroid computation; second, centroids improve resistance to data noise and outliers, as mentioned earlier; third, they are robust against missing values; and lastly, they facilitate training with limited samples, making them suitable for class imbalance learning. Unlike conventional data-level techniques like oversampling or undersampling that modify raw data to handle class imbalance, the FRAME method functions within the latent feature space. This feature rectification in the latent space aims to derive a compact and distinct class representation by continuously adjusting multiple centroids via self-attentive feature interactions (to be introduced in Section II-B2). This strategy alleviates issues related to noise, outliers, and missing values, which are often exacerbated by data-level methods.

FRAME's classifier utilizes centroids obtained from the latent space and applies a distance-based method to assign class membership. Unlike conventional classifiers, which generally work with raw data and often need extra cost-sensitive or reweighting methods to handle class imbalance, FRAME's approach, through its use of latent space centroids, naturally ensures balance and resilience without requiring such modifications.

2) *Centroid Learning Via Self-Attentive Feature Interactions*: To address the issues of data imbalance, noise, and incompleteness in centroid learning, we present an innovative technique that employs self-attentive feature interactions to extract several representative centroids for each class from latent features. In contrast to traditional centroid-based approaches that depend on predefined or fixed criteria (e.g., k-means), our method uses a self-attention mechanism to identify higher-order dependencies among features. This enables FRAME to adaptively learn from complex and imbalanced data distributions, thereby improving its robustness against noise and missing values.

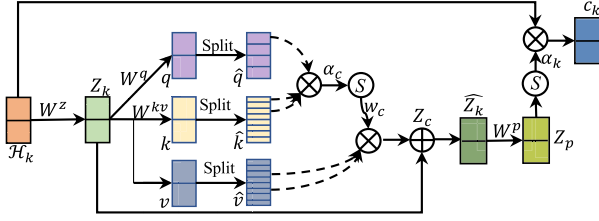


Fig. 2. Illustration of self-attentive learning of centroids.

This method entails a sophisticated two-step process:

1) **Transformation of Raw Data to Latent Features:**

- *Data Input and Transformation:* We commence by transforming the raw data set $\mathcal{X} = [x_1, \dots, x_N] \in \mathbb{R}^{N \times D}$, comprising N samples each with D dimensions, into a set of latent features $\mathcal{H} = [h_1, \dots, h_N] \in \mathbb{R}^{N \times F}$. This transformation is achieved through a parametric mapping $\Phi: \mathcal{X} \rightarrow \mathcal{F}$, which can be instantiated via various models such as neural networks or kernel methods, depending on the data structure.
- *Class-specific Latent Feature Representation:* For each class k , we define $\mathcal{H}_k = [h_1, \dots, h_{N_k}] \in \mathbb{R}^{N_k \times F}$, representing the latent features corresponding to that class.

2) **Self-attentive Feature Interaction for Centroid Generation:**

- *Initial Feature Embedding:* The latent features $\mathcal{H}_k$ of each class are first embedded into a matrix $\mathcal{Z}_k$, using an MLP parameterized by W^z .
- *Projection into Query, Key, and Value Matrices:* Subsequently, $\mathcal{Z}_k$ is projected into query (q), key (k), and value (v) matrices using two separate MLPs (ϕ^{MLP_q} and $\phi^{MLP_{kv}}$), each with its own parameter set.
- *Chunking and Attention Score Calculation:* These matrices are chunked into smaller segments. For each chunk, the module computes dot products between q_c and k_c to derive attention scores α_c , which are normalized to obtain attention weights w_c . These weights signify the relevance of each element within the chunk.
- *Weighted Feature Synthesis and Residual Connection:* The module multiplies w_c with v_c to synthesize weighted features, capturing high-order pairwise interactions. It incorporates a residual connection between these features and the original latent features $\mathcal{Z}_k$, thereby preserving information and mitigating gradient vanishing.
- *Centroid Calculation and Transformation:* The final step involves applying another MLP (ϕ^{MLP_c}) to the enhanced features, thus transforming the self-attention matrix and latent features into a matrix of centroids $\mathbf{c}_k$, representing multiple representative points for each class.

The mathematical formulation of this process is succinctly expressed as $\alpha_k = SA(\mathcal{H}_k)$, $\mathbf{c}_k = \phi^{MLP_c}(\alpha_k \mathcal{H}_k)$, where SA denotes the self-attentive interaction module, and $\mathbf{c}_k \in \mathbb{R}^{n_c \times 1 \times F}$ are the resulting centroids.

Furthermore, to enhance memory efficiency in the self-attention mechanism, a chunking strategy is applied. This

technique involves dividing keys and values into manageable chunks and sequentially computing dot products and updating a running total of weighted values and exponentials, culminating in a normalized attention result. This approach innovatively applies self-attention mechanisms to manage and interpret complex and challenging data characteristics in centroid learning, offering a robust solution to the intrinsic issues of data imbalance and noise.

3) *Distance-Based Classifier:* Conventional training processing first uses a deep neural network to transform raw data into a feature representation. Then, a classifier that uses a single fully connected layer produces a logit vector for each class. Assuming that the feature representation is $h(x) \in \mathbb{R}^{N \times F}$, and the classifier parameters are $W \in \mathbb{R}^{F \times k}$, the classifier operation can be formulated as follows:

$$l = W^T h(x) = [w_1^T h(x); w_2^T h(x); \dots; w_k^T h(x)] \quad (1)$$

where w_i is the weight vector of the i -th class.

Finally, a softmax loss $\mathcal{L}$ is applied to each class to compute the discrepancy between the estimated value and the ground-truth label, which is defined as:

$$\mathcal{L} = \sum_i^k \frac{N_i}{N} \mathcal{L}(D_i) \quad (2)$$

where N_i is the training sample size of the i -th class, N is the latent sample size and D_i is the samples belonging to the i -th class

In (2), if the class imbalance exists and $N_1 \leq N_2 \leq \dots \leq N_k$, we can derive $\mathcal{L}_1 \leq \mathcal{L}_2 \leq \dots \leq \mathcal{L}_k$. The decision boundary will be biased towards the minority class.

Due to the limitations of the softmax loss function, we use a distance-based classifier in this paper and use the classifier to map the samples to the obtained centroids according to their distances. There are several options for determining the probability of a given sample belonging to a class, such as euclidean distance, Cosine similarity, etc. Assume the obtained feature of a sample is denoted as $H_s \in \mathbb{R}^{F \times L}$, the probability belonging to class i is calculated as follows:

$$p(y = i | \mathcal{H}_s) = \frac{\exp(-\|\mathcal{H}_s - \mathbf{c}_i\|^2)}{\sum_k \exp(-\|\mathcal{H}_s - \mathbf{c}_i\|^2)} \quad (3)$$

where $\|\cdot\|$ is the distance function for $\mathcal{H}_s$ and $\mathbf{c}_i$.

4) *Robustness to Class Imbalance:* This section presents a theoretical analysis demonstrating that the proposed loss function, $\mathcal{L}$, exhibits robustness to class imbalance. This robustness stems from the function's dependence on intrinsic feature space properties rather than solely on class size discrepancies.

Consider a dataset $\mathcal{D}$ partitioned into positive (minority) samples, denoted by $\mathcal{P}$, and negative (majority) samples, denoted by $\mathcal{N}$. The cardinalities of these subsets are represented by $N_p = |\mathcal{P}|$ and $N_n = |\mathcal{N}|$, respectively, with the imbalance condition specified as $N_n \gg N_p$. Let h_i represent the latent feature embedding of a data point x_i , and let $\mathbf{c}_0$ and $\mathbf{c}_1$ denote the centroids of the negative and positive classes in the latent space, respectively.

The loss associated with a positive sample x_i is defined as:

$$\mathcal{L}(x_i) = \log(1 + \exp(-d(h_i, \mathbf{c}_0) + d(h_i, \mathbf{c}_1))) \quad (4)$$

where $d(u, v) = |u - v|^2$ represents the squared euclidean distance metric.

Theorem: The overall loss function, $\mathcal{L}$, is primarily governed by the degree of intra-class clustering within the positive class and the inter-class distance between the positive and negative class centroids, demonstrating its resilience to class imbalance.

Proof:

Positive Class Loss: The cumulative loss for all positive instances is given by:

$$\mathcal{L}(\mathcal{P}) = \sum_{i=1}^{N_p} \mathcal{L}(x_i). \quad (5)$$

Lower Bound via Jensen's Inequality: Leveraging the convexity of $\log(1 + \exp(t))$, Jensen's inequality yields:

$$\mathcal{L}(\mathcal{P}) \geq N_p \log \left(1 + \exp \left(\frac{1}{N_p} \sum_{i=1}^{N_p} [-d(h_i, \mathbf{c}_0) + d(h_i, \mathbf{c}_1)] \right) \right). \quad (6)$$

Distance-Based Simplification: Expanding the squared euclidean distances within the exponential term and simplifying:

$$\begin{aligned} \mathcal{L}(\mathcal{P}) \geq N_p \log & \left(1 + \exp \left(-\frac{1}{N_p} \sum_{i=1}^{N_p} \|h_i - \mathbf{c}_0\|^2 \right. \right. \\ & \left. \left. + \frac{1}{N_p} \sum_{i=1}^{N_p} \|h_i - \mathbf{c}_1\|^2 \right) \right). \end{aligned} \quad (7)$$

where $\bar{h} = \frac{1}{N_p} \sum_{i=1}^{N_p} h_i$ represents the mean vector of the positive sample embeddings.

Rewriting this further:

$$\begin{aligned} &= N_p \log \left(1 + \exp \left(-\mathbf{c}_0^T \mathbf{c}_0 + \mathbf{c}_1^T \mathbf{c}_1 \right. \right. \\ & \quad \left. \left. + \frac{2}{N_p} \sum_{i=1}^{N_p} (h_i^T (\mathbf{c}_0 - \mathbf{c}_1)) \right) \right), \\ &= N_p \log \left(1 + \exp \left(-\|\mathbf{c}_0\|^2 + \|\mathbf{c}_1\|^2 \right. \right. \\ & \quad \left. \left. + 2(\bar{h} - \mathbf{c}_0)^T (\mathbf{c}_0 - \mathbf{c}_1) \right) \right). \end{aligned} \quad (8)$$

where the inequality holds if and only if, for all $i \neq j$, we have $-d(h_i, \mathbf{c}_0) + d(h_i, \mathbf{c}_1) = -d(h_j, \mathbf{c}_0) + d(h_j, \mathbf{c}_1)$. Here, $\langle \cdot \rangle$ denotes the inner product, and $\bar{h} = \frac{1}{N_p} \sum_{i=1}^{N_p} h_i$ represents the mean of the latent features of the positive samples. The equality in (8) is achieved at its minimum when $\mathbf{c}_1 = \bar{h}$.

Optimality Condition: The derived lower bound attains its minimum when $\mathbf{c}_1 = \bar{h}$, indicating optimal separation when positive samples cluster tightly around their mean.

Negative Class Loss: Following a similar derivation, the loss for negative samples is bounded by:

$$\mathcal{L}(\mathcal{N}) \geq N_n \log (1 + \exp(-d(\mathbf{c}_1, \mathbf{c}_0))). \quad (9)$$

Overall Loss Behavior: Combining the bounds on positive and negative class losses:

$$\mathcal{L} = \mathcal{L}(\mathcal{P}) + \mathcal{L}(\mathcal{N}) \geq (N_p + N_n) \log (1 + \exp(-d(\mathbf{c}_1, \mathbf{c}_0))). \quad (10)$$

Discussion: The final inequality highlights that the overall loss, $\mathcal{L}$, is primarily dictated by the inter-class distance $d(\mathbf{c}_1, \mathbf{c}_0)$ and the intra-class clustering of positive samples, as evidenced by the influence of $\mathbf{c}_1$ approaching $\bar{h}$. Critically, the dependence on class sizes (N_p and N_n) is less pronounced, signifying the model's inherent robustness to class imbalance. This property is crucial for effective learning in scenarios where class distributions are skewed.

5) Robustness to Missing Values: To analyze the robustness of FRAME against missing values, we consider the Missing Completely at Random (MCAR) mechanism, where the probability of a feature being missing is independent of both observed and unobserved data. Let $x_i \in \mathcal{X}$ be a sample with missing values at indices M_i . We represent the observed features as x_i^o and the missing features as x_i^m . Under MCAR, we have $P(M_i | x_i) = P(M_i)$.

We define the imputation function $\Psi : \mathcal{X} \rightarrow \mathcal{X}'$ that handles missing values by filling them with estimated values. For instance, we can use mean imputation:

$$\Psi(x_i) = (x_i^o, \hat{x}_i^m), \quad \hat{x}_i^m = E[x_i^m] \quad (11)$$

where $E[x_i^m]$ is the mean of the corresponding feature calculated from the complete data.

Impact on Centroid Calculation: Let $\mathbf{c}_k$ be the centroid of class k calculated with complete data, and $\tilde{\mathbf{c}}_k$ be the centroid calculated with missing data handled by Ψ . We analyze the difference between these centroids:

$$\begin{aligned} \|\mathbf{c}_k - \tilde{\mathbf{c}}_k\| &= \left\| \frac{1}{N_k} \sum_{i \in \text{Class } k} \Phi(x_i) - \frac{1}{N_k} \sum_{i \in \text{Class } k} \Phi(\Psi(x_i)) \right\| \\ &\leq \frac{1}{N_k} \sum_{i \in \text{Class } k} \|\Phi(x_i) - \Phi(\Psi(x_i))\| \end{aligned} \quad (12)$$

Assumption 1: The transformation Φ is Lipschitz continuous with constant L :

$$\|\Phi(x) - \Phi(x')\| \leq L \|x - x'\|, \quad \forall x, x' \in \mathcal{X} \quad (13)$$

Using this assumption, we have:

$$\|\mathbf{c}_k - \tilde{\mathbf{c}}_k\| \leq \frac{L}{N_k} \sum_{i \in \text{Class } k} \|x_i - \Psi(x_i)\| \quad (14)$$

Since Ψ only modifies the missing features, we have:

$$\|x_i - \Psi(x_i)\| = \|x_i^m - \hat{x}_i^m\| \quad (15)$$

Assumption 2: The imputation error is bounded:

$$\|x_i^m - \hat{x}_i^m\| \leq B, \quad \forall i \quad (16)$$

Thus, we obtain:

$$\|\mathbf{c}_k - \tilde{\mathbf{c}}_k\| \leq \frac{L}{N_k} \sum_{i \in \text{Class } k} B = LB \quad (17)$$

This demonstrates that the difference between the centroids calculated with complete and missing data is bounded by a constant, indicating a limited impact of missing values on centroid calculation.

Impact on Distance-Based Classifier and Performance: Let $p(y = k|x_i, \mathbf{c})$ be the probability of sample x_i belonging to class k with complete data and centroids $\mathbf{c}$. Let $\tilde{p}(y = k|\Psi(x_i), \tilde{\mathbf{c}})$ be the probability with imputed data $\Psi(x_i)$ and centroids $\tilde{\mathbf{c}}$. We analyze the difference:

$$|p(y = k|x_i, \mathbf{c}) - \tilde{p}(y = k|\Psi(x_i), \tilde{\mathbf{c}})| \quad (18)$$

Using the definition of the distance-based probability (3), we write:

$$p(y = k|x_i, \mathbf{c}) = \frac{\exp(-d(\Phi(x_i), \mathbf{c}_k))}{\sum_j \exp(-d(\Phi(x_i), \mathbf{c}_j))} \quad (19)$$

$$\tilde{p}(y = k|\Psi(x_i), \tilde{\mathbf{c}}) = \frac{\exp(-d(\Phi(\Psi(x_i)), \tilde{\mathbf{c}}_k))}{\sum_j \exp(-d(\Phi(\Psi(x_i)), \tilde{\mathbf{c}}_j))} \quad (20)$$

Assuming the distances $d(\Phi(x_i), \mathbf{c}_j)$ and $d(\Phi(\Psi(x_i)), \tilde{\mathbf{c}}_j)$ are bounded within a range $[d_{min}, d_{max}]$, we utilize the properties of the exponential function and the triangle inequality, along with the Lipschitz continuity of Φ , to derive a bound on the difference in probabilities:

$$\begin{aligned} |p(y = k|x_i, \mathbf{c}) - \tilde{p}(y = k|\Psi(x_i), \tilde{\mathbf{c}})| \\ \leq C \cdot |d(\Phi(x_i), \mathbf{c}_k) - d(\Phi(\Psi(x_i)), \tilde{\mathbf{c}}_k)| \\ \leq C (L \|x_i - \Psi(x_i)\| + \|\mathbf{c}_k - \tilde{\mathbf{c}}_k\|) \\ \leq 2CLB \end{aligned} \quad (21)$$

where $C = \frac{\exp(-d_{min})}{(\exp(-d_{max}))^2}$ is a constant derived from the mean value theorem applied to the exponential function within the specified range.

Connection to Performance: Consider the accuracy of the classifier. Let y_i be the true label of sample x_i . The change in accuracy due to missing values can be bounded by:

$$\begin{aligned} |\text{Accuracy} - \widetilde{\text{Accuracy}}| \\ \leq \frac{1}{N} \sum_{i=1}^N |p(y = y_i|x_i, \mathbf{c}) - \tilde{p}(y = y_i|\Psi(x_i), \tilde{\mathbf{c}})| \\ \leq 2CLB \end{aligned} \quad (22)$$

This shows that the impact of missing values on accuracy is bounded and depends on the Lipschitz constant of Φ , the imputation error bound B , and the constant C . Similar bounds can be derived for other metrics like F1-score or AUC.

C. Applications of FRAME

1) Fully Supervised Learning: To apply the FRAME in a fully supervised learning fashion, we first project the original data into the latent space using an embedding network and then balance and classify the imbalanced features. Fig. 3 illustrates the proposed model FRAME for fully-supervised class imbalance learning. We use dots to represent the data in the original space and other shapes to represent the data in the latent

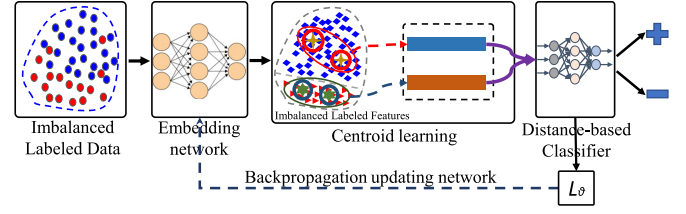


Fig. 3. Illustration of FRAME for fully-supervised class imbalance learning.

space after the transformation. It is important to note that the transformation is applied to both positive and negative samples. The transformation is implemented using the neural network.

Given the raw data $x \in \mathbb{R}^{N \times d}$, we can obtain the transformed features in the latent space as $h \in \mathbb{R}^{N \times F}$ by the embedding network Φ_ω . The transformation is formulated as follows:

$$h_i = \Phi_\omega(x_i), \quad \forall x_i \in \mathcal{N} \cup \mathcal{P}. \quad (23)$$

where ω represents the parameters of the embedding network.

In the fully supervised class imbalance learning, we formulate the centroid learning and classification objectives in a joint framework. The classification loss function for model training is defined using the negative log probability based on the real class label $y \in \{1, 2, \dots, k\}$, as shown in (24). The entire network can be trained by minimizing the classification loss on the training samples using the gradient descent algorithm in an end-to-end manner.

$$\mathcal{L} = - \sum_{(x_k, y_k)} \log(p(y = y_k|x_k, \mathbf{c})) \quad (24)$$

where, (x_k, y_k) represents a sample and $\mathbf{c}$ is the centroid of this sample. The backpropagation algorithm in the neural network will compute the gradient of the loss function and update the model parameters.

2) Semi-Supervised Learning: Semi-supervised learning supports partial labeled data, and it assumes that the classes in both labeled data and unlabeled data are balanced. However, this assumption does not always hold in real-world applications. As a result, performance can be compromised for traditional semi-supervised learning because its decision boundary has been broken. The existing imbalance learning methods, i.e., data-level and algorithm-level methods, are not suited to semi-supervised class learning as they are designed for fully-supervised learning. The most intuitive way of using the unlabeled data is to produce pseudo-labels for them, but this will become a challenge as the imbalance ratios of the labeled and unlabeled data can be different.

To address the above challenge, we extend the FRAME model to support the semi-supervised class imbalance learning. As shown in Fig. 4, the structure of the proposed model for semi-supervised learning is almost the same as that for fully supervised learning, except for the use of unlabeled data to improve centroid learning for the labeled data.

Given the unlabeled data $X_u \in \mathbb{R}^{N_u \times d}$, where N_u is the number of instances of unlabeled data, the latent feature can

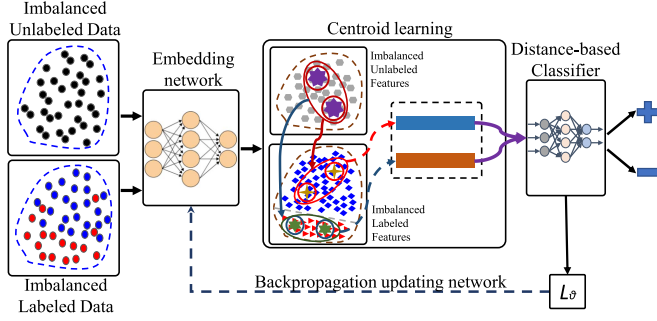


Fig. 4. Illustration of FRAME for semi-supervised class imbalanced learning.

be obtained via the embedding network Φ_ω :

$$\mathcal{H}_u = \Phi_\omega(X_u), \quad (25)$$

Then, the centroids can be learned by:

$$\mathbf{c}_u = \alpha_u \mathcal{H}_u, \quad \alpha_u = SA(\mathcal{H}_u), \quad (26)$$

where $\mathbf{c}_u \in \mathbb{R}^{n_c \times 1 \times F}$ is the prototypes of the unlabeled data. α_u is the weight learned by the self-attentive feature interaction module.

Finally, the centroids of unlabeled data can be used to update the centroids of the labeled data, which is defined as follows:

$$\mathbf{c}_{new}^i = \frac{\alpha \mathbf{c}_l^i + \beta \mathbf{c}_u}{\alpha + \beta}. \quad (27)$$

where $\mathbf{c}_l$ and $\mathbf{c}_u$ are the centroids for the labeled data and the unlabeled data, respectively. α and β are two learnable parameters.

3) *Federated Learning*: As FL is a distributive learning paradigm that aims to protect data privacy, the data-level methods for class imbalance are not suitable for FL due to their need for additional information, such as data distribution. On the contrary, these algorithm-level methods do not require (or less) data information. Therefore, algorithm-level methods are more suitable for class imbalance in FL. Nevertheless, there are lots of hyper-parameters that need to be tuned to ensure each participant can converge at the same speed. Besides the above groups of methods, other researchers also proposed several works. Duan et al. overcome the class imbalance in FL by using some proxy servers [28]. But their method needs each participant to upload the local data distribution, which makes a potential backdoor attack. Wang et al. designed a new loss function for class imbalance in FL. This method can monitor the learning process in the case of class imbalance in FL [29].

Our proposed model can adjust the learning for the class-imbalanced data without the need for data distribution and parameter tuning. In this work, we extend fully supervised learning of the FRAME model with FL in order to support collaborative learning, called *FRAME-FL*. As shown in Fig. 5, FRAME-FL consists of multiple local models and a global model. The local model is the same as that of fully supervised learning. The global model is generated by aggregating all the local models, by a weighted average method, which takes into account the impact of each training sample size. The larger the training sample, the

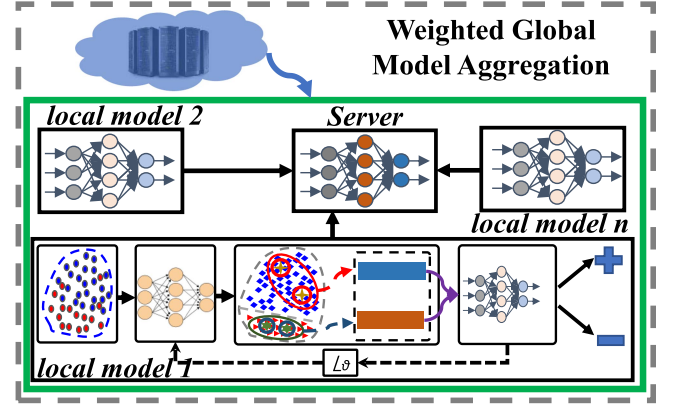


Fig. 5. Illustration of FRAME for class imbalanced federated learning.

higher the weight. The aggregation method is defined as follows:

$$\sum_{i=1}^K \frac{S_i}{K \sum_{i=1}^K S_i} W_{c+1}^i \rightarrow W_{c+1} \quad (28)$$

where, the i -th client model is represented as W_{c+1}^i in the $c+1$ communication round, while the global model in the same round is represented as W_{c+1} . The sample size of the i -th client is denoted by S_i , and the total number of models involved in the learning process is represented by K .

With the FL framework, privacy is maintained through several key steps. Initially, each client trains its own local neural network using its private dataset, which is encoded. Next, each local model computes its gradient descent updates locally and shares only the model parameters with the central server, ensuring that no raw data, which may contain sensitive information, is transmitted. Additionally, clients report the number of training samples to the server to enable weight calculation. Importantly, the central server does not retain any local models or training data from clients; these are discarded immediately after the global model is produced.

IV. EXPERIMENT

In this section, we evaluate our FRAME against current leading methods in fully supervised, semi-supervised, and federated learning across 10 benchmark datasets. Additionally, we examine the model's performance in scenarios with missing data or noise. Detailed information on these 10 benchmark datasets is provided in Table I.

For these 10 benchmark datasets, the model was developed using the Pytorch (v.1.8.0) deep learning framework. All experiments were carried out on a server with a 32GB Tesla v100 GPU. The embedding network consisted of a single-layer Multilayer Perceptron (MLP) with 128 filters. In fully-supervised and semi-supervised learning scenarios, the learning rate for the Adam optimizer was set to $1e-4$. The training process spanned 1000 epochs. For federated learning, each client underwent 100 training epochs, and there were 10 communication rounds between clients and the server. All experiments were repeated five times, employing cross-validation.

TABLE I
SPECIFICATION OF 10 DATASETS

Group	Dataset	IR	# Samples	# Variables
slightly imbalanced	yeast1	2.46	1484	8
	ecoli1	3.36	336	7
	satimage	9.28	6435	36
moderately imbalanced	us_crime	12.29	1994	100
	scene	12.60	2407	294
	ozone_eighthr	14.84	2534	72
highly imbalanced	ozone_level	33.74	2536	72
	poker-8-9_vs_6	58.40	1485	10
severely imbalanced	Protein_homo	111.00	145751	74
	credit_fraud	578.88	284807	31

A. Dataset Description

We commence our evaluation of the proposed method utilizing 10 benchmark datasets dedicated to imbalance learning, sourced from UCI [30] and KEEL [31]. The selection of these datasets aims to ensure both universality and diversity. As illustrated in Table I, these datasets vary in the number of samples and variables and are categorized into four distinct groups based on the imbalance ratio (IR), ranging from slightly imbalanced to severely imbalanced. In this study, ‘slightly imbalanced’ refers to $IR \leq 10$, ‘moderately imbalanced’ represents $10 \leq IR \leq 20$, ‘highly imbalanced’ is defined as $20 \leq IR \leq 100$, and ‘severely imbalanced’ denotes $IR \geq 100$.

B. Evaluation Metric

As the data are imbalanced, the conventional evaluation metric of “accuracy” is no longer appropriate. We therefore use the Matthews correlation coefficient (MCC), defined as follows:

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

$$GMean = \sqrt{\left(\frac{TP}{TP + FN}\right) \times \left(\frac{TN}{TN + FP}\right)} \quad (29)$$

where TP , FP , FN , and TN denote true positives, false positives, false negatives, and true negatives, respectively. MCC is a customized binary classification metric ideal for assessing imbalanced datasets. Furthermore, the AUC (Area Under the Curve) is also utilized for evaluating performance.

C. Experiments on Fully-Supervised Learning

1) *Comparison With State-of-The-Art Methods*: To assess the performance of our proposed model, we carried out a comparative analysis against twelve contemporary baseline models categorized into four groups.

Classical models: **GDO** [4], which incorporates Gaussian distribution into traditional oversampling techniques. **SPE** [6], an ensemble model that uses under-sampling strategies. **DUBE** [1], a recent ensemble model that balances inter-class and intra-class data without relying on extensive distance calculations.

Gradient-boosting methods: **GDBT**, which merges the simplicity of decision trees with the robustness of boosting, achieving high performance across various tasks. **XGBoost** [32], a recent model that integrates XGBoost with focal loss. **LightGBM**, a Gradient Boosting Decision Tree algorithm designed for large datasets, offering faster training speeds while maintaining high prediction accuracy.

Deep loss function: **BalanceSoftmax** [33], an unbiased softmax variant that accounts for label distribution shifts between training and testing. **Class-balance** [34], a focal loss variant that considers the effective number of samples. **LDAM** [35], a loss function that replaces the standard cross-entropy with a label-distribution-aware margin.

Simple classifier: **SVM**, a robust algorithm capable of handling high-dimensional data and determining optimal decision boundaries by maximizing the margin between different classes. **RF**, Random Forest, an ensemble learning method that builds numerous decision trees and outputs the class mode. **DT**, Decision Tree, a supervised learning algorithm that models decisions and their outcomes using a tree-like structure, making predictions by traversing from root to leaf based on input features.

For classical methods, we utilize the default configurations provided. Given the absence of a classifier in the source code of GDO, we substitute it with the AdaBoostClassifier. For gradient-boosting techniques, we incorporate GDBT, XGBoost, and LightGBM as base classifiers within the SPE methods. Regarding loss-based approaches, we withdraw the centroid learning module and replace it with a linear classifier. These loss functions are crafted to enhance model training. The simple classifier also operates with its default settings. Comparative results can be found in Tables II and III.

As shown in Tables II and III, our method outperforms others across all three metrics. The enhanced performance of our model relative to the baselines can be traced to several pivotal factors: 1) **Latent Feature Space Operation**: Unlike methods that manipulate data directly, such as oversampling or undersampling, our approach works within a latent feature space. This choice is beneficial as it proves more effective for data balance compared to altering the original data space. Operating in the latent space offers clear advantages, such as diminished vulnerability to noise and outliers. Moreover, it boosts the model’s ability to define decision boundaries accurately. These benefits cumulatively lead to better model performance. 2) **Centroid-Based Distance Classifier**: Our model employs a classifier based on distances from learned centroids within the latent space, rather than just depending on distances calculated from original features or balancing methods in the data space, like Centroid and SPE. This unique method allows our classifier to utilize feature rectification and balance within the latent space, potentially enhancing performance. 3) **Efficiency Learning**: As opposed to class imbalance learning techniques focused on loss, which introduce penalties and require extensive hyperparameter tuning—a labor-intensive task—our model simplifies the feature balancing process in the latent space. Additionally, it performs classification end-to-end, offering better efficiency and less complexity in achieving a balanced and high-performing model.

TABLE II
PERFORMANCE EVALUATION WITH CLASSICAL AND BOOSTING METHODS IN FULLY-SUPERVISED CLASS IMBALANCE LEARNING

Dataset	Metric	Classical			Gradient-boosting			FRAME
		GDO	SPE	DUBE	GDBT	XGBoost	LightGBM	
yeast1	AUC	0.4409±0.0441	0.4085±0.0429	0.4020±0.0287	0.4611±0.0586	0.4536±0.0495	0.4449±0.0447	0.5052±0.0400
	Gmean	0.5594±0.0217	0.5716±0.0338	0.5484±0.0138	0.6231±0.0437	0.6151±0.0391	0.6015±0.0345	0.6454±0.0266
	MCC	0.3745±0.0785	0.3240±0.0902	0.3094±0.0580	0.4141±0.1149	0.4054±0.0965	0.3895±0.0867	0.4840±0.0564
ecoli1	AUC	0.7240±0.1790	0.6107±0.1653	0.6337±0.1977	0.6758±0.2030	0.6935±0.2070	0.6677±0.2176	0.8270±0.1307
	Gmean	0.8181±0.1138	0.7511±0.1037	0.7520±0.1434	0.7931±0.1306	0.8003±0.1317	0.7805±0.1413	0.8859±0.0858
	MCC	0.7621±0.1574	0.6632±0.1511	0.6745±0.1889	0.7195±0.1897	0.7328±0.1844	0.7040±0.2013	0.8556±0.1077
satImage	AUC	0.1694±0.0325	0.3308±0.1036	0.3498±0.1240	0.3339±0.1100	0.3732±0.1270	0.3669±0.1154	0.4278±0.1459
	Gmean	0.3218±0.0120	0.5429±0.1217	0.5489±0.1287	0.5455±0.1288	0.5775±0.1391	0.5758±0.1217	0.6138±0.1458
	MCC	0.2505±0.0837	0.4722±0.1481	0.4813±0.1680	0.4781±0.1489	0.5189±0.1570	0.5162±0.1385	0.5635±0.1720
us_crime	AUC	0.2236±0.0557	0.2426±0.0545	0.2419±0.0711	0.2899±0.0689	0.3114±0.0987	0.3208±0.1099	0.4257±0.0693
	Gmean	0.4114±0.0721	0.4713±0.0624	0.4436±0.0808	0.5152±0.0736	0.5208±0.1162	0.5292±0.1252	0.6332±0.0562
	MCC	0.3727±0.0741	0.4047±0.0728	0.3936±0.0883	0.4605±0.0831	0.4732±0.1273	0.4814±0.1384	0.5998±0.0612
scene	AUC	0.0973±0.0233	0.1020±0.0160	0.1069±0.0261	0.1267±0.0250	0.1330±0.0339	0.1209±0.0327	0.1821±0.0454
	Gmean	0.2753±0.0095	0.2877±0.0224	0.2800±0.0170	0.3405±0.0359	0.3162±0.0453	0.2951±0.0490	0.3671±0.0693
	MCC	0.1242±0.0666	0.1294±0.0708	0.1399±0.0870	0.2051±0.0898	0.2129±0.0849	0.1887±0.0722	0.3114±0.0741
ozone_eighthr	AUC	0.1471±0.0382	0.1386±0.0327	0.1274±0.0644	0.1765±0.0696	0.2138±0.0496	0.2066±0.0754	0.3091±0.0821
	Gmean	0.3191±0.0526	0.3451±0.0499	0.2968±0.0856	0.3991±0.0900	0.4272±0.0673	0.4099±0.0995	0.5307±0.0751
	MCC	0.2691±0.0804	0.2496±0.0883	0.2114±0.1165	0.3071±0.1400	0.3755±0.0778	0.3539±0.1177	0.4912±0.0825
ozone_level	AUC	0.0880±0.0191	0.0810±0.0121	0.0871±0.0375	0.1011±0.0083	0.1466±0.0788	0.1418±0.0718	0.2783±0.1047
	Gmean	0.2521±0.0392	0.2684±0.0254	0.2554±0.0682	0.3076±0.0121	0.3387±0.1226	0.3365±0.1059	0.5064±0.1062
	MCC	0.2339±0.0426	0.2183±0.0307	0.2189±0.0853	0.2640±0.0151	0.3144±0.1286	0.3121±0.1130	0.4884±0.1116
poker-8-9_vs_6	AUC	0.0562±0.0787	0.0293±0.0081	0.0242±0.0147	0.0193±0.0026	0.0315±0.0293	0.0283±0.0181	0.5396±0.1526
	Gmean	0.1932±0.1270	0.1551±0.0290	0.1438±0.0281	0.1324±0.0054	0.1604±0.0612	0.1485±0.0375	0.7243±0.1016
	MCC	0.0845±0.1799	0.0961±0.0379	0.0293±0.0785	0.0261±0.0296	0.0377±0.1144	0.0471±0.0941	0.7195±0.1037
Protein_homo	AUC	0.0239±0.0047	0.2386±0.0367	0.5546±0.0862	0.6132±0.0817	0.4283±0.0416	0.3779±0.0468	0.6177±0.1038
	Gmean	0.1219±0.0196	0.4857±0.0368	0.7405±0.0603	0.7799±0.0530	0.6525±0.0319	0.6125±0.0380	0.7813±0.0690
	MCC	0.1204±0.0198	0.4788±0.0374	0.7383±0.0606	0.7781±0.0532	0.6488±0.0323	0.6081±0.0384	0.7795±0.0695
creditcard	AUC	0.5036±0.2416	0.0598±0.0172	0.5200±0.1273	0.5512±0.1449	0.4809±0.1714	0.3333±0.1261	0.6871±0.1108
	Gmean	0.6718±0.2271	0.2415±0.0353	0.7152±0.0900	0.7356±0.0988	0.6812±0.1286	0.5669±0.1081	0.8259±0.0677
	MCC	0.6715±0.2272	0.2381±0.0359	0.7147±0.0902	0.7351±0.0991	0.6806±0.1289	0.5658±0.1084	0.8257±0.0678

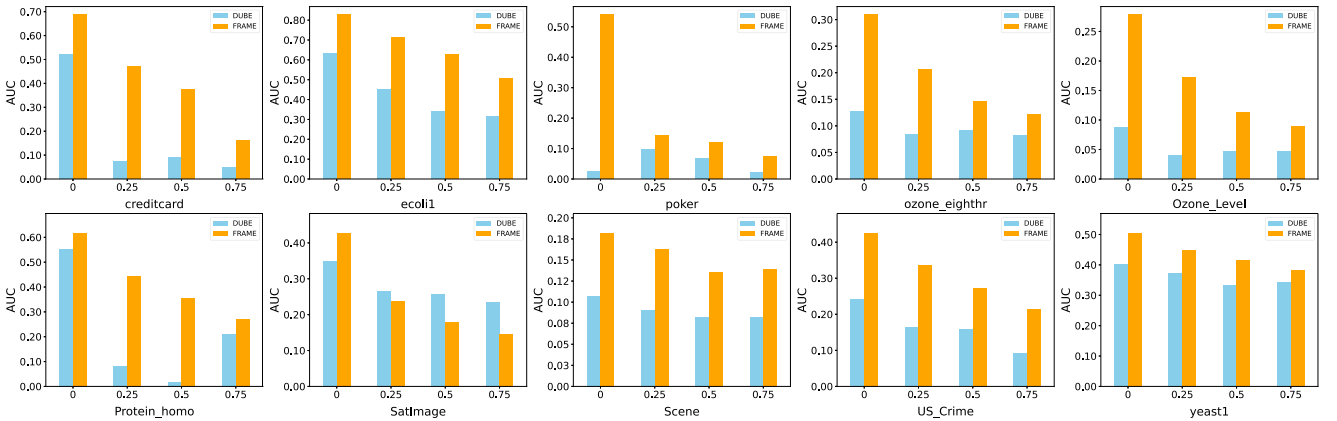


Fig. 6. Performance comparison on missing values for fully-supervised learning.

2) *Performance Evaluation on Missing Values:* As previously mentioned, dealing with missing values is a common challenge in data analysis. These gaps in the data were generated using the MCAR (Missing Completely at Random) model, which posits that the missingness is not related to any observed or unobserved variables [36]. Consequently, we explore the behavior of our proposed model in the presence of missing data. To emulate this, we randomly selected instances from both the training and test datasets and replaced them with zeros.

DUBE was used for comparison due to its superior capability in managing missing data. The tests were conducted with missing data rates of 0%, 25%, 50%, and 75%, and the results are illustrated in Fig. 6.

From the analysis, it is evident that the performance of each approach declines as the proportion of missing data increases. Nevertheless, FRAME demonstrates greater robustness against varying levels of missing data, particularly at higher ratios, across most datasets. Additionally, two noteworthy observations

TABLE III
PERFORMANCE EVALUATION WITH DEEP AND MACHINE LEARNING METHODS IN FULLY-SUPERVISED CLASS IMBALANCE LEARNING

Dataset	Metric	Deep loss function			Simple classifier			FRAME
		BS	CB	LDAM	SVM	RF	DT	
yeast1	AUC	0.4709±0.0312	0.4994±0.0631	0.4826±0.0376	0.4154±0.0462	0.2896±0.0000	0.3665±0.0267	0.5052±0.0400
	Gmean	0.6511±0.0250	0.6368±0.0494	0.6414±0.0393	0.5421±0.0079	0.5377±0.0009	0.5377±0.0009	0.6454±0.0266
	MCC	0.4564±0.0502	0.4684±0.0987	0.4568±0.0630	0.3397±0.0802	0.0003±0.0240	0.2322±0.0680	0.4840±0.0564
ecoli1	AUC	0.6563±0.2162	0.6317±0.2215	0.6947±0.2218	0.6255±0.2106	0.6082±0.1921	0.5864±0.1479	0.8270±0.1307
	Gmean	0.7925±0.1387	0.7607±0.1507	0.8187±0.1357	0.7332±0.1498	0.7224±0.1336	0.7188±0.1154	0.8859±0.0858
	MCC	0.7081±0.2075	0.6675±0.2191	0.7476±0.1958	0.6608±0.1949	0.6381±0.1916	0.6340±0.1496	0.8556±0.1077
satImage	AUC	0.3338±0.1098	0.3026±0.0977	0.2828±0.0705	0.2088±0.1068	0.0973±0.0003	0.3130±0.0911	0.4278±0.1459
	Gmean	0.5501±0.1204	0.5186±0.1041	0.5192±0.0741	0.3936±0.1004	0.3119±0.0005	0.5141±0.1031	0.6138±0.1458
	MCC	0.4790±0.1464	0.4343±0.1470	0.4366±0.0938	0.2635±0.2078	0.0000±0.0000	0.4455±0.1382	0.5635±0.1720
us_crime	AUC	0.3460±0.0785	0.2686±0.0325	0.3150±0.0639	0.2707±0.0719	0.1948±0.0385	0.2099±0.0530	0.4257±0.0693
	Gmean	0.5731±0.0714	0.5089±0.0319	0.5521±0.0608	0.4559±0.0875	0.3592±0.0575	0.4008±0.0754	0.6332±0.0562
	MCC	0.5256±0.0811	0.4470±0.0386	0.4983±0.0714	0.4345±0.0903	0.3416±0.0583	0.3501±0.0801	0.5998±0.0612
scene	AUC	0.1228±0.0286	0.0960±0.0081	0.1233±0.0140	0.0735±0.0009	0.0735±0.0009	0.0848±0.0095	0.1821±0.0454
	Gmean	0.3351±0.0475	0.3010±0.0173	0.3356±0.0269	0.2712±0.0017	0.2712±0.0017	0.2712±0.0017	0.3671±0.0693
	MCC	0.1996±0.0847	0.1386±0.0334	0.2118±0.0410	0.0000±0.0000	0.0000±0.0000	0.0713±0.0442	0.3114±0.0741
ozone_eighthr	AUC	0.1969±0.0717	0.1668±0.0483	0.1779±0.0466	0.0631±0.0000	0.0690±0.0117	0.1007±0.0278	0.3091±0.0821
	Gmean	0.4192±0.0900	0.3866±0.0655	0.4126±0.0605	0.2513±0.0001	0.2513±0.0001	0.2680±0.0211	0.5307±0.0751
	MCC	0.3428±0.1254	0.3000±0.1050	0.3345±0.0829	0.0000±0.0000	0.0343±0.0685	0.1530±0.0947	0.4912±0.0825
ozone_level	AUC	0.0824±0.0240	0.0789±0.0213	0.0849±0.0198	0.0288±0.0010	0.0288±0.0010	0.0495±0.0201	0.2783±0.1047
	Gmean	0.2756±0.0485	0.2642±0.0441	0.2863±0.0363	0.1696±0.0028	0.1696±0.0028	0.1869±0.0357	0.5064±0.1062
	MCC	0.2181±0.0671	0.2085±0.0608	0.2321±0.0493	0.0000±0.0000	0.0000±0.0000	0.1196±0.0605	0.4884±0.1116
poker-8-9_vs_6	AUC	0.1528±0.0085	0.1404±0.0223	0.0946±0.0090	0.0168±0.0000	0.0168±0.0000	0.0362±0.0387	0.5396±0.1526
	Gmean	0.3899±0.0121	0.3698±0.0314	0.3072±0.0151	0.1297±0.0000	0.1297±0.0000	0.1670±0.0746	0.7243±0.1016
	MCC	0.3710±0.0127	0.3499±0.0334	0.2808±0.0170	0.0000±0.0000	0.0000±0.0000	0.0535±0.1280	0.7195±0.1037
Protein_homo	AUC	0.3046±0.1110	0.0580±0.0199	0.3109±0.0804	0.4832±0.1016	0.5819±0.0990	0.5007±0.1002	0.6177±0.1038
	Gmean	0.5402±0.1049	0.2323±0.0436	0.5520±0.0692	0.6873±0.0788	0.7571±0.0702	0.7018±0.0756	0.7813±0.0690
	MCC	0.5344±0.1068	0.2142±0.0468	0.5465±0.0704	0.6857±0.0789	0.7557±0.0704	0.6991±0.0761	0.7795±0.0695
creditcard	AUC	0.0213±0.0125	0.0061±0.0020	0.1096±0.0828	0.4209±0.0976	0.4771±0.0842	0.4338±0.1653	0.6871±0.1108
	Gmean	0.1373±0.0473	0.0737±0.0129	0.3002±0.1387	0.6436±0.0753	0.6874±0.0621	0.6448±0.1324	0.8259±0.0677
	MCC	0.1301±0.0503	0.0621±0.0146	0.2967±0.1409	0.6432±0.0754	0.6870±0.0622	0.6441±0.1328	0.8257±0.0678

can be made. First, the DUBE method exhibits superior robustness compared to FRAME in the SatImage dataset, as FRAME experiences more significant performance drops than DUBE. Second, in datasets like ‘poker’ and ‘Protein homo,’ missing values can actually enhance the performance of DUBE. This improvement is possibly because DUBE directly handles missing values, which may lead to increased significance in subsequent modeling. The robustness of the proposed FRAME method is attributed to its use of an embedding network to capture latent features from the original data, followed by the calculation of feature centroids to classify samples. This process helps minimize the adverse effects of missing values on performance.

3) *Performance Evaluation on Sample Noise*: In our experiment, we focus on analyzing model robustness against typical sample noise [37]. To this end, we injected Gaussian noise into the dataset, holding a mean of 0 while altering variances to create four distinct noise conditions: 1) no noise; 2) variance of 0.1; 3) variance of 0.3; and 4) variance of 0.5. For benchmarking, we consistently used DUBE, and the results are depicted in Fig. 7.

The results indicate that both our method and the DUBE method exhibit a downward trend in performance as the noise rate rises. However, our method demonstrates a slightly less steep decline compared to the DUBE method. Notably, when a moderate noise level (variance = 0.1) is introduced, the model’s performance does not deteriorate but actually improves slightly for datasets like ozone_eighthr and satimage. The significant

drop in DUBE’s performance could be attributed to its under-sampling approach, which attempts to learn potential relationships from the original data; adding more noise to the test data alters its structure, leading to a sharp decline. While our model’s performance also diminishes with increased noise, the extent of the decrease is smaller, potentially due to discrepancies in the distance calculations between noisy test data samples and the training data centroids, causing only a minor reduction.

4) *Ablation and Sensitivity Analysis*: We assess the impact of the feature refinement module within our model through an ablation study. Specifically, we substitute this module with a linear classifier, resulting in a modified model referred to as “FRAME w/o C”. The outcomes are depicted in Fig. 8. The findings indicate a significant decline in performance upon the removal of the feature refinement module, thereby highlighting its role in enhancing imbalance learning capabilities.

Next, we proceed to assess the impact of varying the number of centers and chunks within the feature refinement module. This evaluation is accomplished through detailed sensitivity analysis by altering these critical parameters and executing extensive experiments on the 10 datasets. The findings from these experiments are clearly presented in Fig. 9. According to the results, the optimal performance is observed when the number of chunks is 256 and the number of centers is 2 for most datasets. The rationale behind incorporating multiple centers in the feature refinement module is to tackle the issue of latent space

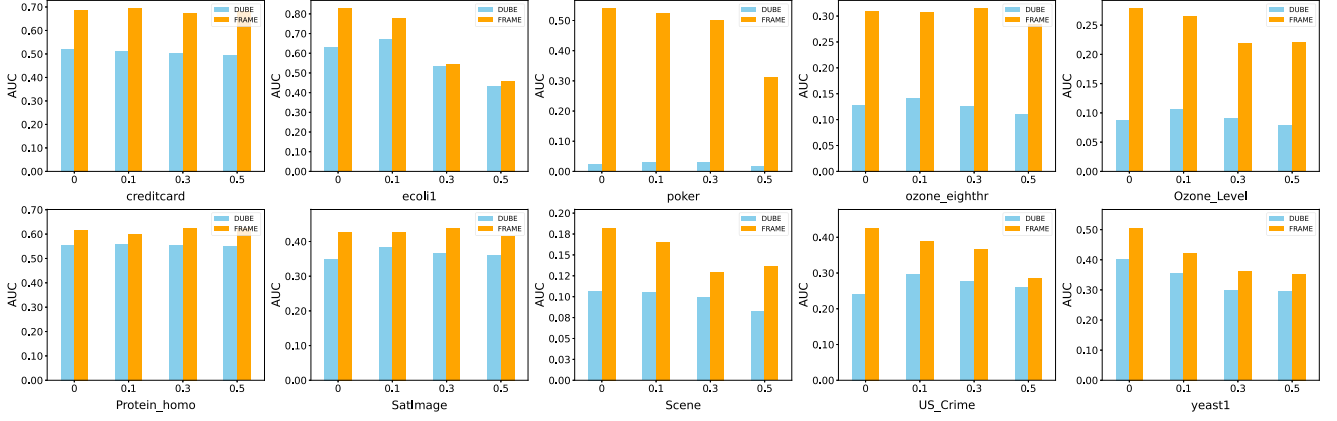


Fig. 7. Performance comparison on sample noise for fully-supervised learning.

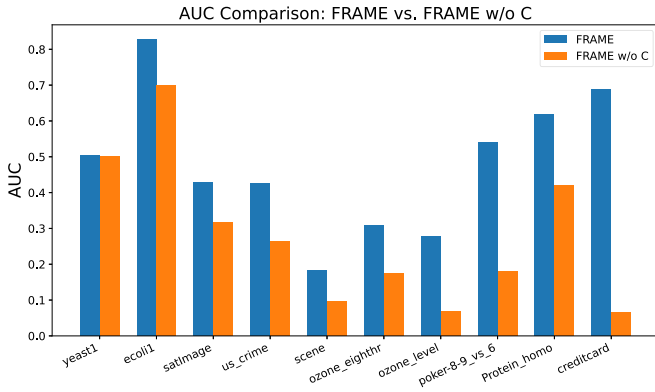


Fig. 8. Ablation study on fully-supervised learning.

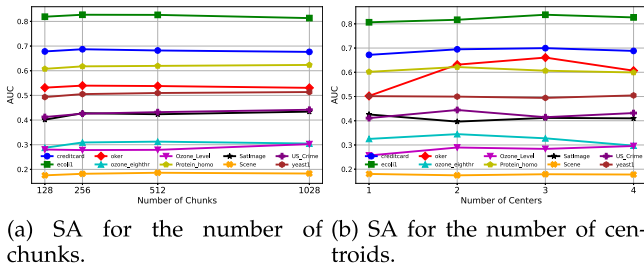


Fig. 9. Sensitivity analysis on fully-supervised learning.

separation. However, it is important to highlight that, in cases where latent features are already well-clustered, the addition of multiple centers may unintentionally degrade performance.

D. Experiments on Semi-Supervised Learning

In this study, we selected five leading contemporary models for performance benchmarking: **II** [38], which enhances deep learning with minimal samples by combining cross-entropy and MSE loss; **PL** (Pseudo Label) [39], a popular semi-supervised learning method that begins with both labeled and unlabeled data, then selects the highest-confidence unlabeled samples; **MT** (Mean Teacher) [40], which improves test accuracy while reducing label dependency by averaging model weights; **VAT**

(Virtual Adversarial Training) [41], a label-free approach using virtual adversarial loss for computational efficiency; and **ICT** (Interpolation Consistency Training) [42], which steers decision boundaries towards low-data-density regions through interpolation for computational efficiency. In our experiments, we used the same embedding network for our model and selected baselines, focusing on their ability to learn from unlabeled samples. We split the data into an 80/20 train/test ratio, with just 10% of the training data labeled and the rest unlabeled. Default settings are employed for the methods, and we fine-tuned our model's hyperparameters. Results are reported in Table IV. The obtained results highlight that our model excels, achieving the highest performance in all datasets.

The four methods, II, PL, VAT, and ICT, utilize unique loss functions in their respective training procedures designed to tackle class imbalance and generate high-quality pseudo-labels for future semi-supervised learning tasks. In contrast, MT adopts a mutual learning framework involving both teacher and student models to address class imbalance effectively and capitalize on unlabeled data. Although these methods can deliver competitive outcomes, their main focus lies in managing class imbalance and leveraging the potential of unlabeled samples. Our approach, however, aggregates centroids from both labeled and unlabeled data to maximize the information contained within unlabeled samples. This comprehensive integration allows us to fully exploit the inherent patterns in the unlabeled data. Additionally, our model offers flexibility in fine-tuning the weights given to labeled and unlabeled data, promoting optimal performance. The careful use of unlabeled data through our multi-center learning design and the ability to adjust data weights make our model a more adaptable and efficient solution for enhancing performance in semi-supervised learning contexts.

E. Experiments on Federated Learning

We choose the following five state-of-the-art models as the baselines for comparison. The clients of the baselines use the embedding network for feature learning but differ in using loss functions to address the class imbalance problem, which are listed as follows: **FL_LADM**: LADM loss function; **FL_CB**: CB loss function; **FL_FOCAL**: Focal loss function;

TABLE IV
PERFORMANCE EVALUATION WITH STATE-OF-THE-ART METHODS IN SEMI-SUPERVISED CLASS IMBALANCE LEARNING

Datasets	Metric	ICT	MT	PI	PL	VAT	FRAME
yeast1	AUC	0.3501±0.0507	0.4544±0.0401	0.4561±0.0400	0.4717±0.0326	0.4582±0.0455	0.4905±0.0324
	Gmean	0.5524±0.0244	0.5874±0.0291	0.5959±0.0251	0.5912±0.0339	0.5957±0.0357	0.6343±0.0250
	MCC	0.1783±0.1330	0.4060±0.0375	0.4102±0.0251	0.4287±0.0428	0.4132±0.0348	0.4656±0.0307
ecoli1	AUC	0.5498±0.1356	0.5302±0.1081	0.6565±0.0886	0.6241±0.1089	0.6420±0.0789	0.7915±0.0425
	Gmean	0.7293±0.0956	0.7701±0.0608	0.8047±0.0552	0.7729±0.0646	0.7924±0.0482	0.8782±0.0202
	MCC	0.6338±0.1198	0.6916±0.0737	0.7405±0.0661	0.6978±0.0804	0.7231±0.0576	0.8415±0.0238
satImage	AUC	0.1618±0.0373	0.1378±0.0097	0.2800±0.0160	0.1402±0.0119	0.2818±0.0149	0.3857±0.0162
	Gmean	0.3625±0.0370	0.3404±0.0226	0.5024±0.0146	0.3420±0.0255	0.5024±0.0142	0.5820±0.0144
	MCC	0.2166±0.1036	0.1835±0.0283	0.4294±0.0178	0.1887±0.0313	0.4307±0.0141	0.5447±0.0136
us_crime	AUC	0.1238±0.0252	0.1293±0.0130	0.1953±0.0372	0.3016±0.0384	0.2086±0.0331	0.3635±0.0866
	Gmean	0.3190±0.0410	0.3330±0.0284	0.4263±0.0400	0.5202±0.0348	0.4438±0.0332	0.5693±0.0750
	MCC	0.1601±0.1017	0.1941±0.0662	0.3372±0.0411	0.4699±0.0348	0.3598±0.0306	0.5316±0.0785
scene	AUC	0.1059±0.0154	0.1197±0.0141	0.1174±0.0128	0.1242±0.0220	0.1174±0.0117	0.1649±0.0308
	Gmean	0.2954±0.0214	0.3048±0.0075	0.3138±0.0234	0.3071±0.0380	0.3189±0.0229	0.3361±0.0308
	MCC	0.1423±0.0541	0.1610±0.0397	0.1821±0.0448	0.1922±0.0561	0.1836±0.0510	0.2858±0.0565
ozone_eighthr	AUC	0.1150±0.0193	0.1125±0.0162	0.1303±0.0214	0.1974±0.0398	0.1379±0.0214	0.2765±0.0476
	Gmean	0.3174±0.0337	0.2994±0.0136	0.3456±0.0281	0.4052±0.0475	0.3595±0.0259	0.4924±0.0476
	MCC	0.2279±0.0416	0.1952±0.0132	0.2656±0.0272	0.3654±0.0562	0.2821±0.0249	0.4664±0.0517
ozone_level	AUC	0.0575±0.0159	0.0964±0.0316	0.0616±0.0201	0.1183±0.0644	0.0658±0.0240	0.2508±0.0928
	Gmean	0.2218±0.0404	0.2117±0.0211	0.2351±0.0446	0.3118±0.1020	0.2424±0.0494	0.4732±0.0961
	MCC	0.1633±0.0457	0.1419±0.0201	0.1738±0.0592	0.2755±0.1183	0.1833±0.0644	0.4648±0.0985
poker-8-9_vs_6	AUC	0.0374±0.0104	0.2807±0.1860	0.0501±0.0228	0.2308±0.1147	0.0561±0.0264	0.2846±0.0636
	Gmean	0.1820±0.0301	0.2583±0.1606	0.2133±0.0493	0.4450±0.1458	0.2282±0.0516	0.5195±0.0627
	MCC	0.1375±0.0398	0.2432±0.1661	0.1767±0.0511	0.4398±0.1497	0.1930±0.0568	0.5159±0.0633
Protein_homo	AUC	0.0196±0.0078	0.1684±0.0181	0.0123±0.0024	0.1579±0.0154	0.2495±0.0156	0.5920±0.0348
	Gmean	0.1311±0.0292	0.4067±0.0217	0.1070±0.0096	0.3937±0.0192	0.4973±0.0157	0.7669±0.0231
	MCC	0.0821±0.0553	0.3983±0.0222	0.0462±0.0287	0.3848±0.0198	0.4909±0.0159	0.7654±0.0231
creditcard	AUC	0.0058±0.0030	0.1076±0.0626	0.0023±0.0007	0.0491±0.0164	0.0884±0.0090	0.4649±0.1266
	Gmean	0.0706±0.0215	0.2134±0.0229	0.0453±0.0068	0.2170±0.0376	0.2963±0.0152	0.6738±0.1019
	MCC	0.0537±0.0312	0.2101±0.0229	0.0187±0.0120	0.2137±0.0379	0.2938±0.0151	0.6732±0.1023

FL_BaSoftmax: BalanceSoftmax loss function; and **FL_Ratio**: a ratio-based loss function [29].

Assuming three clients in the federated learning framework, we utilize FedAvg as the aggregation mechanism for generating the global model [43]. The Adam optimizer’s learning rate is set to $1e-4$, with each client training for 100 epochs and communication rounds set to 10. The data is split in an 80/20 ratio for training and testing, respectively, and the training data is further randomly divided into three parts for training each client model. Table V presents the results. On average, our model surpasses the baseline models, excelling in 8 out of 10 datasets. Notably, the recently proposed FL_Ratio model does not perform as well as anticipated. The key to our method’s superior performance over the state-of-the-art approaches is its straightforward and effective strategy for addressing class imbalance. Specifically, our method uses centroids to directly combat class imbalance, tackling the issue at its core. In contrast, the baseline methods depend on their loss functions’ inherent learning capabilities, which might lack direct and explicit means to address class imbalance. This difference gives our model a performance edge.

F. Discussions

In this paper, we present FRAME, a groundbreaking latent feature rectification module designed for addressing class imbalance learning. FRAME equalizes feature distribution within the latent space, thereby boosting classification performance across different learning paradigms. We also provide theoretical

support that validates FRAME’s effectiveness in resolving class imbalance issues. Our assessment of 10 public binary imbalanced datasets under fully-supervised, semi-supervised, and federated learning paradigms indicates that FRAME surpasses the performance of current state-of-the-art methods. Additionally, the test results on the Long-Tailed CIFAR-100 (**detailed in the Supplementary material**) also highlight the superiority of the proposed model in multi-class imbalance classification.

Our contributions to class imbalance learning are multifaceted. First, we introduce a self-attentive feature interaction module that independently learns multiple centroids per class from latent features, capturing each sample’s significance and diversity while reducing the influence of noise, outliers, or missing data. Second, we design a distance-based classifier focused on these learned centroids, eliminating the need for algorithmic tweaks or extra hyperparameters. Finally, we demonstrate FRAME’s versatility by incorporating it into various learning paradigms, proving its relevance to numerous class imbalance situations.

However, our work has limitations. First, the methodology currently employs a trial-and-error technique to ascertain the optimal number of centroids for each class. Prospective investigations might delve into the development of adaptive methodologies, tailored to optimize the centroid count in alignment with the data’s distribution or the specific requirements of the task at hand. Second, our model’s efficacy has not been evaluated in environments characterized by a vast array of categories, exemplified by datasets akin to ImageNet. In such intricate contexts,

TABLE V
PERFORMANCE EVALUATION WITH STATE-OF-THE-ART METHODS IN FEDERATED CLASS IMBALANCE LEARNING

Datasets	Metric	CB	Focal	LDAM	BS	FL Ratio	FRAME
yeast1	AUC	0.5082±0.0359	0.5121±0.0313	0.5085±0.0311	0.5130±0.0356	0.4413±0.0298	0.4879±0.0280
	Gmean	0.6316±0.0252	0.6265±0.0263	0.6268±0.0276	0.6280±0.0273	0.6124±0.0164	0.6242±0.0253
	MCC	0.4803±0.0386	0.4851±0.0375	0.4816±0.0365	0.4870±0.0356	0.4098±0.0392	0.4558±0.0425
ecoli1	AUC	0.7800±0.0869	0.7786±0.0805	0.7852±0.0869	0.7805±0.0795	0.7652±0.0264	0.8056±0.0346
	Gmean	0.8721±0.0571	0.8707±0.0534	0.8751±0.0545	0.8728±0.0514	0.8434±0.0123	0.8796±0.0278
	MCC	0.8316±0.0737	0.8309±0.0671	0.8362±0.0693	0.8325±0.0663	0.7643±0.0857	0.8440±0.0436
satImage	AUC	0.4741±0.0236	0.4695±0.0293	0.4787±0.0351	0.4762±0.0263	0.4536±0.0242	0.4698±0.0241
	Gmean	0.6614±0.0173	0.6579±0.0214	0.6644±0.0265	0.6631±0.0190	0.6213±0.0156	0.6584±0.0154
	MCC	0.6284±0.0196	0.6243±0.0243	0.6321±0.0290	0.6303±0.0215	0.5825±0.0437	0.6117±0.0141
us_crime	AUC	0.3036±0.0384	0.3022±0.0251	0.3021±0.0334	0.3078±0.0333	0.3642±0.0152	0.4054±0.0385
	Gmean	0.5254±0.0376	0.5274±0.0235	0.5238±0.0290	0.5299±0.0276	0.5765±0.0246	0.6144±0.0294
	MCC	0.4728±0.0380	0.4734±0.0179	0.4715±0.0273	0.4785±0.0253	0.5482±0.0365	0.5778±0.0255
scene	AUC	0.1197±0.0173	0.1203±0.0164	0.1194±0.0167	0.1193±0.0157	0.1577±0.0145	0.1828±0.0204
	Gmean	0.3275±0.0290	0.3309±0.0273	0.3268±0.0274	0.3277±0.0263	0.3356±0.0124	0.3674±0.0207
	MCC	0.1986±0.0245	0.2019±0.0274	0.1973±0.0243	0.1986±0.0219	0.2418±0.0260	0.3139±0.0153
ozone_eighthr	AUC	0.2795±0.0335	0.2744±0.0280	0.2833±0.0325	0.2800±0.0294	0.2652±0.0225	0.3103±0.0562
	Gmean	0.5082±0.0353	0.5018±0.0310	0.5099±0.0357	0.5080±0.0316	0.4882±0.0213	0.5357±0.0550
	MCC	0.4711±0.0413	0.4653±0.0349	0.4744±0.0414	0.4716±0.0369	0.4420±0.0555	0.5024±0.0618
ozone_level	AUC	0.1056±0.0315	0.1013±0.0309	0.1042±0.0317	0.1048±0.0290	0.1008±0.0435	0.2318±0.0372
	Gmean	0.3028±0.0512	0.2961±0.0515	0.3013±0.0521	0.3024±0.0490	0.2834±0.0452	0.4660±0.0426
	MCC	0.2685±0.0547	0.2613±0.0544	0.2659±0.0559	0.2682±0.0512	0.2033±0.0676	0.4491±0.0433
poker-8-9_vs_6	AUC	0.6457±0.1953	0.6574±0.1882	0.6379±0.1636	0.6292±0.2029	0.6126±0.0876	0.6801±0.2366
	Gmean	0.7906±0.1190	0.7919±0.1146	0.7875±0.1025	0.7787±0.1249	0.7506±0.0250	0.7938±0.1615
	MCC	0.7881±0.1206	0.7886±0.1161	0.7850±0.1039	0.7760±0.1265	0.6963±0.1183	0.7913±0.1636
Protein_homo	AUC	0.7250±0.0280	0.7281±0.0254	0.7273±0.0257	0.7165±0.0102	0.7170±0.0121	0.7283±0.0306
	Gmean	0.8502±0.0167	0.8520±0.0150	0.8515±0.0152	0.8411±0.0142	0.8432±0.0152	0.8612±0.0181
	MCC	0.8490±0.0167	0.8508±0.0151	0.8503±0.0153	0.8429±0.0131	0.8380±0.0143	0.8514±0.0182
creditcard	AUC	0.6017±0.0229	0.5979±0.0277	0.6020±0.0229	0.6032±0.0252	0.5987±0.0145	0.6153±0.0136
	Gmean	0.7753±0.0149	0.7726±0.0181	0.7754±0.0150	0.7762±0.0163	0.7624±0.0126	0.7793±0.0125
	MCC	0.7749±0.0149	0.7723±0.0181	0.7750±0.0150	0.7758±0.0163	0.7213±0.0132	0.7834±0.0131

there is a propensity for the centroids assigned to each category to converge, potentially diminishing the model's effectiveness. Future scholarly efforts could be directed towards enhancing the precision of centroid identification methods. Lastly, our approach utilizes the euclidean distance metric to gauge the similarity between features and centroids. Future research could explore the integration of more sophisticated distance metrics or similarity functions, such as the Mahalanobis distance or kernel-based approaches, to potentially elevate the model's performance.

V. CONCLUSION AND FUTURE WORK

In this paper, we proposed FRAME, a generic framework for dealing with a variety of classical class imbalance problems. In this paper, we uncovered the key element for dealing with class imbalance classification problems, i.e., how to accurately estimate the centroids of a data set. We proposed to map the data features from the original space to the latent space using an attention mechanism, then estimate the centroids in the latent space, and finally balance the classes in the latent space to obtain better accuracy. For better usability, we extended FRAME to support fully supervised learning, semi-supervised learning and federated learning conditions. For evaluation, we conducted extensive experiments on up to 10 different binary-class datasets

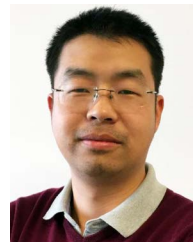
against different baselines to validate the generality and effectiveness of the proposed framework under the above three learning conditions. Furthermore, we evaluated the robustness of the framework to data noise and missing values. All the experimental results showed that FRAME is a promising unified solution for handling a variety of classical class imbalance learning problems. It can be adapted to different neural network structures and requires less training data but can achieve results.

There are a number of interesting directions for future work, among others, including label noise learning, open set recognition, and incomplete information learning. In addition, distributional inconsistency problems such as domain adaptation and transfer learning are also worthy of further investigation.

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